

AC01226

13th April

[Int]

[1, 2, 3, 4]

[[Int]]

[[1, 2, 3], [4, 5]]

reverse :: [Int] → [Int]

reverse [] = []

reverse (x:xs) = (reverse xs) ++ [x]

(y:ys) → (z:zs)

reverse [1, 2, 3] = [3, 2, 1]

Suppose we have a list of integers and we want to obtain another list of integers by squaring each integer of the first list

$\text{Square} :: \text{Int} \rightarrow \text{Int}$
 $\text{Square } x = x * x$

$[1, 2, 3] \rightarrow [1, 4, 9]$

$\text{SquareAll} :: [\text{Int}] \rightarrow [\text{Int}]$

$\text{SquareAll } [] = []$

$\text{SquareAll } (x:xs) = (\text{Square } x) : (\text{SquareAll } xs)$

built-in
function in
Haskell

$$\underline{\underline{\text{map}}} \quad f \quad [x_0, x_1, \dots, x_k] \\ = [f(x_0), f(x_1), \dots, f(x_k)]$$

$$\text{squareAll} :: [Int] \rightarrow [Int]$$

$$\text{squareAll} \quad \underline{\underline{=}} \quad \text{map} \quad \underline{\underline{\text{square}}} \quad \underline{\underline{=}} \\ \text{squareAll} \quad \underline{\underline{=}} \quad \text{map square}$$

$\text{listlengths} :: [[\text{Int}]] \rightarrow [\text{Int}]$

$\text{listlengths } [] = []$

$\text{listlengths } (a:xs) = (\text{length } a) : (\text{listlengths } xs)$

$\text{listlength } l = \text{map length } l$

listlength $=$ map length

filter just like the way map operates

filter p l retains those elements in l for which p evaluates to True

for example,

isEven :: Int $\rightarrow$ Bool

isEven $x = (\text{mod } x \ 2 == 0)$

getEvens :: [Int] $\rightarrow$ [Int]

$\left\{ \begin{array}{l} \text{getEvens } [] = [] \\ \text{getEvens } (x:xs) \end{array} \right. \mid \begin{array}{l} (\text{isEven } x) = x : (\text{getEvens } xs) \\ \text{otherwise} = \text{getEvens } xs \end{array}$

can be written as
filter isEven

list comprehension

$$\{ n^2 \mid n \in \{1, 2, 3, 4, 5\}, \text{odd}(n) \}$$

$$[n^2 \mid n \leftarrow [1, 2, 3, 4, 5], \text{isOdd } n]$$

divisors n

$\{1, \dots, n\}$
 $\downarrow$
 m

m divides n

$\text{divisors } n = [m \mid m \leftarrow [1..n],$
 $\text{mod } n \ m == 0]$

$\text{isPrime } n :: \text{Int} \rightarrow \text{Bool}$

$\text{isPrime } n = (\text{divisors } n == [1, n])$

Sorting

Insertion Sort

$\text{insertionSort} :: [\text{Int}] \rightarrow [\text{Int}]$

$\text{insertionSort } [] = []$

$\text{insertionSort } (x:xs) = \text{insert } x (\text{insertionSort } xs)$

where

$\text{insert } x [] = [x]$

$\text{insert } x (y:ys) \mid (x < y) = x:(y:ys)$

$\mid \text{otherwise} = y:(\text{insert } x \text{ } ys)$

Mergesort

$\text{merge} :: [\text{Int}] \rightarrow [\text{Int}] \rightarrow [\text{Int}]$

$\text{merge } [] \ l = l$

$\text{merge } \ l \ [] = l$

$\text{merge } (x:xs) (y:ys) \mid (x < y) = x : (\text{merge } xs (y:ys))$

$\mid \text{otherwise} = y : (\text{merge } (x:xs) ys)$

$\text{mergesort} :: [\text{Int}] \rightarrow [\text{Int}]$

$\text{mergesort } [] = []$

$\text{mergesort } [x] = [x]$

$\text{mergesort } l = \text{merge } (\text{mergesort } (\text{fsthalf } l))$
 $\quad \quad \quad (\text{mergesort } (\text{sndhalf } l))$

where

$n = \text{div } (\text{length } l) \ 2$

$\text{fsthalf } l = \text{take } n \ l$

$\text{sndhalf } l = \text{drop } n \ l$

~~Quicksort:~~

quicksort $:: [Int] \rightarrow [Int]$

quicksort $[] = []$

quicksort $(a:as) = (\text{quicksort } \text{smaller})$

$++ [a] ++$

$(\text{quicksort } \text{larger})$

where

$\text{smaller} = [y \mid y \in as, y \leq a]$

$\text{larger} = [y \mid y \in as, y > a]$

no need
to merge
unlike in
mergesort

$\text{length} :: \underline{\underline{[Int]}} \rightarrow \underline{\underline{Int}}$

$\text{length} :: \underline{\underline{[Bool]}} \rightarrow \underline{\underline{Int}}$

$\text{length} :: \underline{\underline{[True, False, True]}}$

$\text{length} :: \underline{\underline{[0.5, 0.2, 0.0]}}$

$\text{length} [] = 0$

$\text{length} (x:xs) = 1 + (\text{length } xs)$

~~length~~

$\text{length} :: \underline{\underline{[Float]}} \rightarrow \underline{\underline{Int}}$

$\text{length} :: \underline{\underline{[a]}} \rightarrow \underline{\underline{Int}}$

type variable



quicksort $\therefore [int] \rightarrow [int] \checkmark$

$[float] \rightarrow [float] \checkmark$

$[bool] \rightarrow [bool] \times$

$[a] \rightarrow [a]$ under
some
conditions

$[a]$

$\rightarrow$ such that
I can compare